Project 8 – Multi-link aeronautical system

Consider an aeronautical communication system composed of N aircraft and K independent data links connecting the aircraft to a ground station. Each aircraft generates one message with a fixed size of L bytes every T seconds, where T is a RV to be described later.

Data links have the same bandwidth of C bytes per second and are shared by all aircraft. Each data link can carry one packet at a time according to a First-In-First-Out (FIFO) policy.

When an aircraft generates a new message, the latter is transmitted according to either of the following policies:

a) *Single-path transmission*: the aircraft selects one of the K data links randomly, where each data link has the same probability of being chosen. The message is transmitted as one packet of $B = L$ bytes.
b) *Multi-path transmission*: the message is divided into K equal-sized packets, and one packet is transmitted through each data link. Each transmitted packet has size $B = L/K$ bytes. The ground station can reconstruct the original packet only after all its segments have been received.

For each data link, the transmission time for a packet of B bytes is $S = D + B/C$ seconds, where D is an additional per-packet service time due to channel-access procedures, and is a RV to be described later. Each transmission draws an independent value of D. A data link remains occupied for the entire transmission time.

Evaluate at least the end-to-end delay of messages under both transmission policies and for various values of N and K.

At least the following scenarios must be evaluated:

- Exponential distribution of both T and D.
- Exponential distribution of T and lognormal distribution of D with the same mean as in the previous case.

In all cases, it is up to the team to calibrate the scenarios so that meaningful results are obtained.

The project rules, required deliverables, and selected frequently asked questions are provided below.

Rules and instructions

The team must **model and analyze the assigned system using discrete-event simulation**, according to the simulation workflow presented during classes. The simulator code must be implemented using the OMNeT++ framework.
The assignment defines the operating rules of the system, the minimum performance measures, and the minimum set of policies/factors/scenarios that must be studied. The team is free to add metrics or experiments when they are useful to better understand the system behavior or provide additional, relevant insights.
The outcome of the project should clarify *how* the system behaves and *why* the system behaves that way, by providing *meaningful insights* about the system. A "meaningful insight" can be an important trend, a specific tradeoff, a non-trivial behavior, or any other information that allows the audience to deepen their understanding of the system. For instance, claiming that "the response time of the system increases with the service time" is obvious and is therefore unlikely to be a meaningful result.

Project deliverables

Each group must submit the following deliverables:

1. **Report**: a PDF document describing the simulation study (more details below).
2. **Simulator code**: a ZIP archive including all source files and simulation configurations needed to run the experiments. Binary/executable files and simulation result files *must not* be included.
3. **Presentation**: up to 10 PowerPoint slides with a concise account of the system, the analysis objectives, modeling, simulation configurations and main results.

Submission and project discussion

Submission procedure. Upload all deliverables to the group's dedicated MS Teams channel, then post a message (tag the instructor using '@') in the same channel clearly stating that the group intends to submit the project. The instructor will acknowledge receipt. Submission is final: revised versions cannot be uploaded after the submission has been acknowledged.
Timing. There is no fixed project deadline. However, students *cannot* take the written exam before the project deliverables have been submitted. A student planning to take a particular written exam session must therefore submit the project before the official date of that written exam.
Project discussion. All members of a group must present the project together on the same occasion, even if they plan to take the written or oral examinations in different sessions. The discussion will take place before the oral examination, on a date communicated by the instructor.

Recommended report structure

The report should be self-contained: a reader should be able to understand the model and assess the conclusions without inspecting the source code. A suitable, non-mandatory structure is:

4. **System description.** Summarize the assigned system and its operating rules in your own words.
5. **Objective and performance indicators.** State the goal of the analysis and list the metrics that will be measured to gain the corresponding insights. Remember that "evaluating metric X" is not a SMART objective.
6. **System model.** Describe how the system has been modeled (i.e., its entities and how they interact), along with any assumptions you made. Where needed, motivate modeling choices and assumptions.
7. **Simulator design.** Briefly explain the structure of the implemented OMNeT++ simulator, without turning the report into a line-by-line code description.
8. **Verification and validation.** Present test cases and observed outcomes. Validation against theoretical models (if possible) is appreciated, but not mandatory.
9. **Experimental design.** List scenarios, factors, and parameter values used in the simulation campaign.
10. **Results.** Present the evidence obtained from the simulations using tables and figures, providing convincing explanations.
11. **Conclusions.** Provide answers to the analysis objectives, summarizing the obtained results.

Frequently Asked Questions

Q: Is the project mandatory?
A: Yes, it is.

Q: Is the project individual or group work?
A: The project must be completed in groups of three students.

Q: I do not like my teammates. Can I change groups?
A: No, you cannot.

Q: I do not like my project assignment. Can I receive a different one?
A: No, you cannot.

Q: One member of our group has dropped out of the project. What should we do?
A: The remaining group members must complete the project. Please inform the instructor as soon as possible.

Q: How should we communicate with the instructor?
A: Preferably use the group's dedicated Teams channel created by the instructor within the PECSN team. Alternatively, just send an email copying all group members and specify your group number.

Q: In which language should the report and presentation be prepared?
A: The report must be prepared in English. The presentation can be given in either English or Italian.

Q: If we have questions about the assignment or need help, can we meet with the instructor?
A: Certainly. Send a message in your dedicated Teams channel to schedule a meeting with the instructor. Tag the instructor using "@" so that they receive a notification.

Q: Must every group member understand the entire project?
A: Yes. Every group member is expected to know the system model, simulator code, experimental design, and results. During the project discussion, the instructors may ask individual member questions about any part of the work.

Q: What does the project discussion consist of?
A: Each group has 15 minutes to present its work to the instructors using the submitted slides. Students should organize the presentation so that every member has an opportunity to speak.
The instructors may ask questions about the project, including the implemented code and the content of the report.

Q: Will the project be graded?
A: The project will receive a qualitative assessment, such as "ottimo", "molto buono", ..., or "insufficiente". Instructors will evaluate the work based on (i) the report, (ii) the simulator code, and (iii) the presentation and the answers provided during the discussion. This assessment will contribute to the final grade, together with the grades obtained in the written and oral examinations.

Q: If the project assessment is not satisfactory, can we revise our work and resubmit the deliverables?
A: No. Submission and discussion are final.

Q: If the project assessment is not satisfactory, can we be assigned a different project?
A: No. Submission and discussion are final.

Q: Does the submitted project remain valid indefinitely?
A: Yes. Once the project has been submitted, students may take the exam in any subsequent examination session.